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AHMED SOLIMAN, Qatar University, Doha, Qatar

YALDA ZAFARI-GHADIM, Qatar University, Doha, Qatar

ESSAM A RASHED, University of Hyogo, Kobe, Hyogo, Japan

MOHAMED ABDALLA MABROK, Qatar University, Doha, Qatar

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Efficient Attention-guided Transformer for brain stroke segmentation

Ahmed Soliman
Department of Mathematics and Statistics, College of Arts and Sciences, Qatar University
Doha, Qatar

Yalda Zafari-Ghadim
Department of Mathematics and Statistics, College of Arts and Sciences, Qatar University
Doha, Qatar

Essam A. Rashed
Graduate School of Information Science, University of Hyogo
Kobe, Japan

Mohamed Mabrok
Department of Mathematics and Statistics, College of Arts and Sciences, Qatar University
Doha, Qatar

ABSTRACT

Ischemic brain stroke occurs when a thrombus blocks a brain artery leading to a regional damage of brain due to lack of normal blood flow. Identification and diagnosis of stroke requires quick processing of medical image such as MRI. Segmentation of brain strokes is known to be a challenging task due to variability in the size and location of the stroke. In this paper, we examine a Transformer-based deep learning architecture, called Dual Attention-guided efficient Transformer (DAE-Former) for several stroke benchmarks. For efficient evaluation, several loss function combinations were examined along with dipping the original architecture, resulting in the Enhanced DAE-Former. The proposed modification leads to marginal improvement in the segmentation accuracy, especially evident in the results obtained from the ATLAS v2.0 dataset. In contrast, the performance improvements were not as pronounced with the smaller ISLES22 dataset. Both the original and Enhanced DAE-Former architectures performed similarly on this dataset. This outcome suggests that the enhanced model are more apparent in larger datasets, where the increased complexity of the architecture can be fully leveraged.

CCS CONCEPTS

• **Computing methodologies → Image segmentation**.

KEYWORDS

Transformers, stroke segmentation, medical imaging, deep learning.

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1 INTRODUCTION

Brain stroke, a sudden loss of brain function due to disrupted blood supply, is a critical health issue [19]. Rapid diagnosis is vital as therapy timing significantly affects recovery. Medical imaging offers insight into the body's anatomy and function [8], aiding in disease diagnosis, treatment tracking, and clinical applications like radiotherapy and surgery [1]. Stroke segmentation in images such as CT scans or MRI helps isolate affected brain areas. Performing the segmentation by medical professionals is subjective to human error and time-consuming. However, employing automatic models face challenges due to variability in stroke shapes and positional differences across patients for accurate segmentation [16].

Convolutional Neural Networks (CNNs) excel in computer vision tasks like image classification and medical segmentation, including stroke segmentation [11, 13]. Yet, they may struggle with long-range spatial relationships crucial for complex stroke patterns [23]. Transformers [6], a recent architecture, revolutionizing natural language and computer vision, excel in capturing global context through self-attention mechanism. This makes them ideal for tasks like stroke segmentation, where understanding overall brain structure is essential.

In this paper, an assessment study of the original DAE-Former architecture [4], which has demonstrated promising results in the context of medical image segmentation, is conducted. Moreover, we propose a model improvement to amplify its segmentation capabilities. Specifically, a fusion loss function is employed with architecture re-engineered through additional dual attention blocks to enhance the network's ability to capture intricate features.

2 RELATED WORK

Considering the recent development in the field, the segmentation architectures for medical images can be categorized into three types: CNN-based, Transformers-based, and Hybrid (a combination of CNNs and Transformers). In the following, a brief discussion on related work is presented.

2.1 CNN-based architectures

CNN has played a pivotal role in the field of stroke segmentation, with efforts concentrated on refining the U-shaped design to boost accuracy [4, 17, 18]. For instance, a supervised U-Net network with five parallel layers was devised to segment stroke lesions in FLAIR

MRI images [12]. An unsupervised feature learning approach employing a stacked sparse autoencoder (SSAE) was introduced to autonomously learn relevant features from brain MR images [20]. Additionally, a hybrid architecture merging U-Net with Fractal-Net showed enhanced accuracy by removing unnecessary segments, and employing a voting system improved the results further [14]. The authors in [5] introduced a spatiotemporal segmentation model focusing on symmetry, capturing micro-perfusion dynamics in the brain while generating a static segmentation map for evaluating infarct cores. Their suggested PerfU-Net utilized an attention module on the skip connections to align the dimensions of the encoder and decoder components.

2.2 Transformers-based architectures

Transformers have shown to be extremely successful in various applications, including medical imaging. However, the development of pure Transformer architectures for segmentation has been limited until now. Swin-Unet [3], a convolution-free architecture for medical image segmentation, divides images into non-overlapping regions, utilizing each patch as a token for deep feature representation learning. Its decoder upsamples context features with a patch-expanding layer and fuses them with encoder multi-scale features via skip connections to restore spatial resolution. Dual Attention-Guided Efficient Transformer (DAE-Former) is another Transformer-based architecture for medical image segmentation [2]. The design combined efficient attention [24] with transpose attention [7] in a block called the efficient dual attention block to improve the model's ability to capture more contextual features.

2.3 Hybrid architectures

Researchers are actively addressing limitations in both CNNs and Transformers. While CNNs struggle with capturing long-range dependencies and global context, Transformers face scalability issues with quadratic time and space complexity concerning sequence length. Hybrid models or innovative architectures that combine the strengths of both are being explored to achieve more accurate and robust segmentation, particularly in stroke segmentation. One such approach is the Hybrid UNet and Transformer (HUT) network, which was designed with two concurrent stages: one based on UNet and the other on Transformers [21]. The intermediate features extracted by the UNet's CNN encoder were fed into the Transformer block to improve segmentation accuracy by enhancing attention and relationship modeling between 3D brain volume voxel patches. A multi-encoder Transformer (METrans) was developed by to address the U-Net-based network's failure to model long-range contextual interactions [22]. Unlike typical segmentation approaches, four encoders with different scales were investigated to extract multi-scale features. The feature maps were then sent to a Transformer to be utilized for global feature modeling.

3 DAE-FORMER

3.1 Original DAE-Former

Dual attention-guided efficient Transformer (DAE-Former) is a Transformer-based architecture specifically crafted for medical image segmentation, drawing inspiration from the U-Net framework [2]. Characterized by dual attention blocks embedded in both the encoder and decoder, alongside cross-attention blocks within the skip connections, the model aims to enhance segmentation outcomes. The dual attention blocks leverage efficient attention techniques [24], effectively reducing computational complexity from quadratic to linear. Additionally, the incorporation of Transpose attention [7] in these blocks facilitates channel attention. This architectural choice is informed by empirical evidence suggesting that the amalgamation of spatial and channel attention improves the model's capacity to capture a richer set of contextual features [9]. Notably, this design was tailored for segmenting human organs in the Synapse dataset, encompassing nine distinct classes (eight organs and the background). The authors of DAE-Former reported that their architecture outperforms state-of-the-art methods on multi-organ cardiac and skin lesion segmentation datasets without requiring pre-training, achieving a dice score of 0.8263 on the Synapse dataset and 0.9147 on the ISIC2018 dataset. The model's architecture is illustrated in Figure 1.

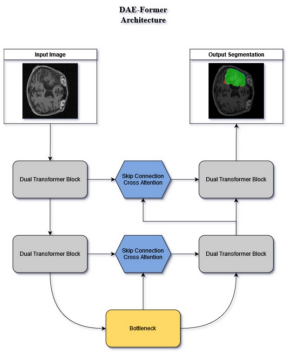

Figure 1: Original DAE-Former architecture that consists of 2× dual attention block in encoder and decoder, and 2× skip connections cross-attention [2].

3.2 Enhanced DAE-Former (EDAE-Former)

An enhanced version of the DAE-Former architecture is proposed with additional features such as including four dual attention encoder blocks and four dual attention decoder blocks, as compared to the original three blocks for each. Moreover, an additional skip connection has been added, resulting in a total of three skip connections instead of the original two. These changes were made to improve the model's ability to grasp intricate patterns and dependencies in medical imaging data. Additionally, we have made important refinements to the loss functions employed in the original implementation Equation 1.

$$\text{Loss} = \alpha \cdot L_{\text{dice}} + \beta \cdot L_{\text{ce}}, \tag{1}$$

where $\alpha = 0.6$ and $\beta = 0.4$ are hyper-parameters. Departing from the initial combination of multi-class dice loss and binary cross-entropy, we have adopted a binary dice loss and binary cross-entropy with logits loss as well. Notably, a weight of 10 has been assigned to the stroke class within the loss function, underscoring its significance in the segmentation task. This tailored adjustment

aims to address the unique challenges associated with stroke segmentation and aligns the model's learning objectives more closely with the specific characteristics of the target class.

In experimental studies, we systematically explored various weighting combinations for the loss function components. After extensive trials, we determined that the most effective balance was achieved with a weight of 0.6 for the Dice loss and 0.4 for the cross-entropy loss. This specific configuration consistently yielded the best results in our tests, striking an optimal balance between the benefits of each loss component for our application.

4 DATASETS

4.1 ISLES dataset

The Ischemic Stroke Lesion Segmentation (ISLES) datasets are publicly released as a challenge related to stroke lesion segmentation. In EDAE-Former, we utilized the ISLES22 version dataset [10] with samples shown in Figure 2. This dataset contains 400 MRI cases from multiple vendors, including 250 cases that are publicly accessible and 150 cases that remain private to the owners. Each case includes three different modalities (DWI, ADC, and FLAIR). We have split the 250 available cases into training and testing sets for our experiments. Specifically, 80% of these cases (the first 200) were allocated for training, while the remaining 20% (the last 50 cases) were reserved for testing. We also extended our training strategy by incorporating data from all three spatial axes—Axial, Coronal, and Sagittal.

4.2 ATLAS dataset

The Anatomical Tracings of Lesions After Stroke (ATLAS) datasets are available in two versions, namely 1.2 and 2.0. These datasets include T1-weighted MRI cases along with their corresponding lesion masks. Version 1 contains a total of 304 cases, while version 2 is more extensive, encompassing 955 cases. We have employed version 2.0 [15] with samples shown in Figure 3 as it has large enough cases needed for the training. Among these 955 cases, 655 T1-weighted MRI scans are publicly accessible. We randomly partitioned the data into a training set, constituting 75% of the cases (491 cases), and a testing set, comprising the remaining 25% (164 cases). Additionally, we have exclusively used the Axial plane for both the training and testing phases.

5 EXPERIMENTAL SETUP

All the results reported in this section were obtained with a workstation equipped with NVIDIA A100 GPU with 80GB memory. The training configuration included a batch size of 24, an initial learning rate of 0.05, a weight decay of 0.0001, an input image size of 224×224 pixels, and the Stochastic Gradient Descent (SGD) optimizer.

6 EVALUATION METRICS

6.1 Dice Score

The Dice score (DS) serves as the primary evaluation metric in this study, offering a quantitative measure of segmentation accuracy. DS is a statistical metric used to assess the similarity between two sets. It measures the overlap between the predicted segmentation and the ground truth, producing a value between 0 and 1, where 1 indicates perfect segmentation alignment and 0 indicates no overlapping. The DS between sets A and B is defined as:

$$\mathrm{DS} = \frac{2|A \cap B|}{|A| + |B|} = \frac{2 \times \mathrm{TP}}{2 \times \mathrm{TP} + \mathrm{FP} + \mathrm{FN}}, \tag{2}$$

where TP, FP, and FN represent true positive, false positive, and false negative, respectively. One challenge we faced in utilizing DS arises from the variety of methods available for its calculation. This variability complicates result comparison across different models and studies. To comprehensively address this issue, three distinct approaches for calculating DS were employed here, each providing unique insights into the performance of our segmentation model.

6.1.1 Slice-based DS. The first method involves calculating DS for each slice independently. By obtaining DS for each slice, we gain a more detailed understanding of the segmentation performance at a single slice level. After calculating these individual scores, the mean of all DS is computed, providing an average measure of segmentation accuracy across the entire dataset.

6.1.2 Volume-based DS. In the second method, which is a three-dimensional (3D) voxel-based calculation, DS is calculated for each volume separately. Subsequently, the mean of these scores is taken across all cases.

6.1.3 Aggregated DS. Moving on to the third method, instead of calculating and then averaging DS for each volume/slice, TP, FP, and FN values across all cases were aggregated first. Following this aggregation, DS is computed just once at the end of the process.

6.2 HD95

The mean Hausdorff Distance (mean HD) provides a measure of the similarity between two sets of points. This method works the same as the Volume-base DS method where the HD95 is calculated for each volume separately. Subsequently, the mean of these scores is taken across all cases.

7 RESULTS

In this study, we adapted the original architecture to address our binary semantic segmentation problem. The corresponding results for ISLES22, and ATLAS v2.0 datasets are reported for training of 100 epochs are shown in Table 1 row (1 & 5). The original architecture provided excellent results, surpassing the state-of-the-art benchmarks for both the ISLES22 and ATLAS v2.0 datasets. The efficacy of the proposed EDAE-Former architecture is substantiated by achieving a higher DS compared to state-of-the-art results on the ATLAS v2.0 dataset. Several modifications were made to the trains to enhance model results. Rows (2) and (6) show the results for the DAE-Former architecture with modified losses: weighted binary cross entropy with weights of 10 and a binary loss function. Rows (3) and (7) display the results for the EDAE-Former with cross-entropy loss and multi-class dice loss function. Rows (4) and (8) present the results for the EDAE-Former with modified losses: weighted binary cross entropy with weights of 10 and a binary loss function. As indicated in Table 1, combining the two modifications (Deeper DAE-Former with modified losses) yields the highest results for the ATLAS v2.0 dataset, while the results for ISLES22 were almost similar. The improved segmentation performance on the ATLAS v2.0

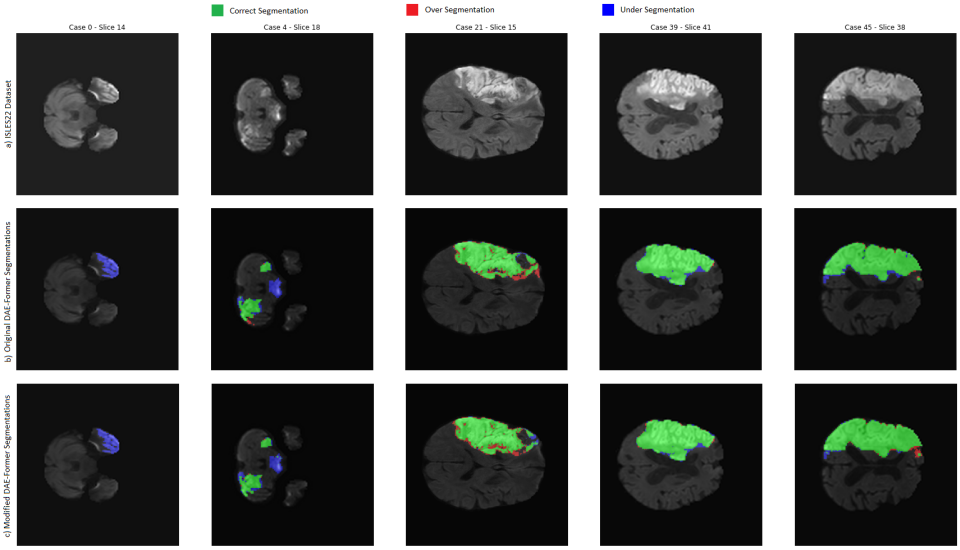

Figure 2: a) Sample of ISLES22 dataset b) ISLES22 segmentation's results for the original DAE-Former [2] architecture c) ISLES22 segmentation's results EDAE-Former architecture

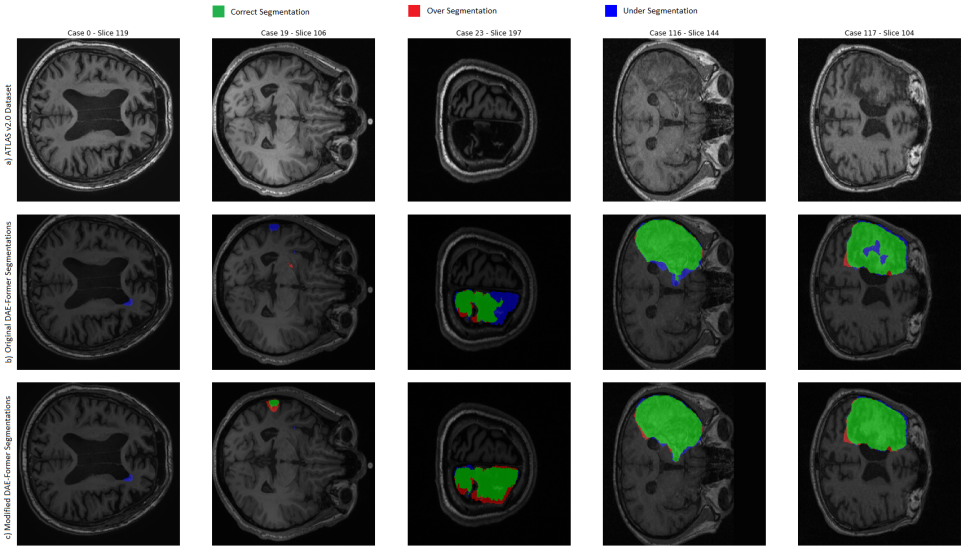

Figure 3: a) Sample of ATLAS v2.0 dataset b) ATLAS v2.0 segmentation's results for the original DAE-Former [2] architecture c) ATLAS v2.0 segmentation's results of EDAE-Former architecture

dataset underscores the effectiveness of our proposed modifications in enhancing the DAE-Former architecture for stroke segmentation tasks on large datasets. As shown in Figure 3, the segmentation results of the modified architecture Figure 3(c) is more accurate than the original architecture segmentations Figure 3(b). In contrast to DS achieved by the original architecture, the modifications made did not yield noticeable improvements in the segmentation results for the ISLES22 dataset as shown in Figure 2(b,c). This lack of improvement can be attributed to the relatively small size of the dataset. The limited diversity and quantity of data may have constrained the effectiveness of the modifications, hindering the model's ability to generalize and capture subtle patterns within the dataset.

DISCUSSION AND CONCLUSION

In this paper, we have evaluated the DAE-Former architecture and presented the EDAE-Former for brain ischemic stroke segmentation, showing improved performance on the ATLAS v2.0 dataset but similar results on the smaller ISLES22 dataset. A key challenge

Table 1: Dice score and Mean HD95 for different datasets on DAE-Former and EDAE-Former architectures. The best results for each dataset are shown in bold.

Dataset	Model	Loss function	Slice-based DS	Volume-based DS	Aggregated DS	Mean HD95
ISLES22	DAE-Former	CE + Dice	0.70364	**0.70321**	**0.84965**	**13.41640**
		WBCE + Binary Dice	0.67930	0.68694	0.84284	24.87958
	EDAE-Former	CE + Dice	**0.71761**	0.69763	0.84202	20.89619
		WBCE + Binary Dice	0.68096	0.69782	0.84799	16.76305
ATLAS v2.0	DAE-Former	CE + Dice	**0.77443**	0.40943	0.78276	97.65244
		WBCE + Binary Dice	0.73544	0.39351	0.76849	**93.56208**
	EDAE-Former	CE + Dice	0.73090	0.37328	0.75371	94.68903
		WBCE + Binary Dice	0.77423	**0.42757**	**0.79462**	97.40636

faced was the lack of standardized methods for calculating DS, leading to inconsistencies in model evaluation. Additionally, the limited availability of datasets for stroke segmentation, restricted to only ISLES22 and ATLAS v2.0, posed constraints on our study. Future work will focus on reducing the false positive ratio to enhance the model's clinical applicability. Despite these challenges, the assisted DAE-Former marks a step forward in stroke segmentation, highlighting the need for standardized evaluation methods and more diverse datasets in medical imaging research.

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